

DEEPSEA COMMUNICATIONS TRAINING SESSION

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I. INTRODUCTION

"Welcome to our third meeting. We'll follow the same dynamics as previous sessions:

Please keep microphones muted

Submit questions via Zoom chat

I'll coordinate with Mauricio to address questions

Today's agenda focuses on communications protocols for Deepsea controllers (2-2.5 hours)."

II. TRAINING AGENDA

Dipsy Controller Communication Resources

Modbus Protocol

855 Converter (USB-Ethernet)

Web Scouting Mute (74/86 Series)

J1939 Communication Protocol

Configurable CAN Instrumentation

SNMP Protocol

DSC Webnet & Gateway Configuration

Note: Some topics target advanced developers, while others cover basic user configurations.

III. PRODUCT OVERVIEW

A. Deepsea Communication Products:

DSC Webnet & smartphone app

Remote displays (7" 8004 / 10" 8003 MKII)

Gateways:

892 (SNMP protocol)

891 (Ethernet)

855 USB-Ethernet converter

B. Included Software:

Skada (free Modbus-compatible software)

Configuration Suite (includes 8005 software for 73/74/86 series)

IV. LIVE DEMONSTRATIONS

A. 855 Converter Setup

Compatibility:

Works with all Dipsy controllers

Supports LAN/WAN connections

Key Features:

Web browser configuration

Built-in SCADA system accessible via IP

Allows multiple remote accesses simultaneously

Limitations:

Embedded Web SCADA not available for controllers with native Ethernet ports

Live demo shows web interface configuration and Modbus TCP/IP integration.

B. Modbus Protocol Deep Dive

Implementation:

Uses Gencom port addressing

Supports RS-485 (recommended) and RS-232

Practical Example:

Reading generator power values:

$\text{Register} = (\text{Page_hex} \times 256) + \text{Offset}$

Example: Page 0x166 (358 dec), Offset 0 → Register 42496

Demo with Modscan software showing real-time battery voltage reading

Compatibility Notes:

All controllers support configurable Gencom pages

Works with third-party equipment (e.g., ASCO ATS)

C. J1939/CAN Protocol

Key Concepts:

29-bit identifier structure (Priority-PGN-Source Address)

SPN (Suspect Parameter Number) system

Configuration Demo:

Setting up oil pressure transmission:

Priority: 0xC8

PGN: 0xFEE9

Source Address: 0x27

Application:

Electric engine monitoring

Fuel level sensors

D. SNMP Protocol Configuration

Security Advantages vs Modbus:

IP whitelisting (max 2 managers)

Dedicated ports for notifications

Read/write passwords

Setup Process:

Load MIB file from Deepsea website

Configure community strings (public/private)

Version 2 only

Embedded Web SCADA:

Direct browser access to controller interface

Demo shows live engine monitoring with password-protected controls

V. DSC WEBNET SYSTEM

A. Core Features

Real-time monitoring (4-sec refresh)

GPS location tracking

Multi-user access (unlimited accounts)

Automated reports/alerts (SMS/email)

B. Gateway Setup

Models:

890 MK-II (GSM/Ethernet/GPS)

891 (Ethernet only)

Configuration Steps:

Set Gateway ID (critical for account linking)

Data resolution settings (High/Medium/Low)

Historic snapshot intervals

Cloud Integration:

Free service (no subscription)

Mobile apps for iOS/Android

C. Account Management Demo

Creating user groups with custom privileges

Configuring event triggers (alarms → SMS notifications)

Generating automated PDF/Excel reports

Live walkthrough of a client account monitoring 400+ generators

VI. Q&A SUMMARY

Q1: Can 335 controller communicate simultaneously via 890 gateway and 855 converter?

A: Yes - modern processors handle parallel communications without priority issues.

Q2: What happens if firmware update interrupts Webnet connection?

A: Controller auto-reconnects unless IP address changes.

Q3: Are MIB files standardized?

A: No - each controller has unique MIB template matching its parameters.

Q4: Webnet licensing costs?

A: Completely free after gateway purchase.

VII. CLOSING REMARKS

"Thank you for attending this 2.5-hour session. Next week we'll cover EMI series controllers.

Special thanks to Mauricio and José Pablo for coordination.

For follow-up questions:

Technical issues: contact Setecom

Deepsea configurations: support@deepsea.com

Good afternoon!"